

Equation Sheet: Electrostatics

Intro Physics II • PHYS 235 (calc) • Knight ch. 22–25

fundamental = know cold (your starting points); **derived** = a line or two from the fundamentals; **bonus** = not required — might appear as an extra-credit problem only.

Symbols and constants

q, Q charge (C)	Φ electric flux (N·m ² /C)	U potential energy (J)
$\vec{E}$ electric field (N/C = V/m)	λ, η, ρ line / area / volume charge density	V potential (V = J/C)
r, d distance / separation (m)	A area	e 1.60×10^{-19} C
ϵ_0 8.85×10^{-12} C ² /(N·m ²),	$\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9$ N·m ² /C ²	
$\vec{r}_{12}$ separation vector, <i>points from charge 1 to charge 2</i> : $\vec{r}_{12} = \vec{r}_2 - \vec{r}_1$; $\hat{r}_{12} = \vec{r}_{12}/r_{12}$		

1. Coulomb's law & the electric field (Knight 22.4–22.5, 23.1–23.2)

fundamental

Coulomb's law, vector form
 ($\hat{r}_{12}$ points from charge 1 to charge 2; force *on* 2, *due to* 1; repulsive for like signs, $F \propto 1/r^2$)

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

Definition of the electric field

$$\vec{F} = q\vec{E}$$

Superposition of forces

$$\vec{F}_{\text{net}} = \sum_i \vec{F}_i$$

derived

Field of a point charge
 ($\hat{r}$ points from the charge to the field point)

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

Superposition of fields

$$\vec{E}_{\text{net}} = \sum_i \vec{E}_i$$

Coulomb's law, alternate vector form

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} q_1 q_2 \frac{\vec{r}_{12}}{r_{12}^3}$$

2. Continuous charge distributions (Knight 23.3–23.6)

fundamental

Chop into pieces, add up (integrate)

$$\vec{E} = \int d\vec{E}, \quad dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$$

Charge densities

$$\lambda = \frac{Q}{L}, \quad \eta = \frac{Q}{A}, \quad \rho = \frac{Q}{V}$$

derived

Infinite line of charge (distance r)
 (direction: radially outward from the line, for $\lambda > 0$)

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

Infinite plane / sheet of charge
 (direction: perpendicular to the sheet, away from it on both sides, for $\eta > 0$)

$$E = \frac{\eta}{2\epsilon_0} \quad (\text{uniform — no } r)$$

Ring of charge Q , radius R , on axis at z
 (direction: along the axis, away from the ring for $Q > 0$)

$$E = \frac{1}{4\pi\epsilon_0} \frac{Qz}{(z^2 + R^2)^{3/2}}$$

Disk of charge Q , radius R , on axis at z
 ($\eta = Q/\pi R^2$; direction: along the axis, away from the disk for $Q > 0$; $R \rightarrow \infty$ recovers the infinite sheet; $z \gg R$ recovers a point charge)

$$E = \frac{\eta}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right)$$

3. Electric flux & Gauss's law (Knight 24.1–24.3)

fundamental

Electric flux through a surface

$$\Phi = \oint \vec{E} \cdot d\vec{A}$$

Gauss's law

(any closed surface; only the *enclosed* charge matters)

$$\Phi = \frac{Q_{\text{enc}}}{\epsilon_0}$$

derived

Uniform field through a flat area

$$\Phi = EA \cos \theta \quad (\theta \text{ from the normal})$$

Charge *outside* a closed surface

$$\Phi_{\text{net}} = 0$$

Sphere / spherical shell, total charge Q
($r > R$)
($\hat{r}$ points from the center outward)

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r} \quad (\text{same as a point charge})$$

4. Conductors in equilibrium (Knight 24.4–24.6)

fundamental

Field inside the conducting material

$$\vec{E} = 0$$

Excess charge

all on the outer surface

Field at the outer surface

$$\perp \text{ to the surface,} \quad E = \frac{\eta}{\epsilon_0}$$

derived

Empty cavity inside a conductor

$$\vec{E} = 0 \text{ in the cavity} \quad (\text{Faraday shielding})$$

Charge $+q$ placed in the cavity

$-q$ on the inner wall, $+q$ on the outer surface

5. Electric potential energy & potential (Knight 25.1–25.7)

fundamental

Two point charges

(keep the signs; $U \rightarrow 0$ as $r_{12} \rightarrow \infty$; one power of r)

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

Energy conservation

$$K_i + U_i = K_f + U_f, \quad W_{\text{elec}} = -\Delta U$$

Definition of potential

$$U = qV \quad (1 \text{ V} = 1 \text{ J/C})$$

Point charge

(scalar; keep the sign; $V \rightarrow 0$ as $r \rightarrow \infty$)

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}, \quad V_{\text{net}} = \sum_i \frac{1}{4\pi\epsilon_0} \frac{q_i}{r_i}$$

derived

Three or more charges: sum over every *pair*

(the work to assemble the whole arrangement — order doesn't matter)

$$U_{\text{total}} = \frac{1}{4\pi\epsilon_0} \sum_{\text{pairs } (i,j)} \frac{q_i q_j}{r_{ij}}$$

Equipotentials

$$\perp \text{ field lines; } \vec{E} \text{ points toward decreasing } V$$

Potential in a uniform field

(d measured along $\vec{E}$, from initial to final point)

$$V_f - V_i = -Ed$$

6. Electric dipoles (Knight — confirm section #)

fundamental

Dipole moment ($\vec{d}$ points from $-q$ to $+q$)	$\vec{p} = q\vec{d}$
Torque on a dipole in a uniform field	$\vec{\tau} = \vec{p} \times \vec{E}$
Potential energy of a dipole in a uniform field (most stable, i.e. minimum U , when $\vec{p} \parallel \vec{E}$)	$U = -\vec{p} \cdot \vec{E}$

7. Doing the calculus: continuous distributions & Gauss's law (Knight 23.3–23.6, 24.1–24.3, 25.4)

fundamental

Potential of a continuous distribution — chop, add up (V is a scalar: no components, no cancellation to track)	$V = \int dV, \quad dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$
Evaluating a Gaussian-surface integral (pick a surface where E is constant and either $\parallel$ or $\perp$ to $d\vec{A}$ everywhere)	$\oint \vec{E} \cdot d\vec{A} = E \oint dA = EA \Rightarrow E = \frac{Q_{\text{enc}}}{\epsilon_0 A}$

derived

Ring of charge Q , radius R , on axis at z	$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{z^2 + R^2}}$
Disk of charge Q , radius R , on axis at z ($\eta = Q/\pi R^2$)	$V = \frac{\eta}{2\epsilon_0} (\sqrt{z^2 + R^2} - z)$

8. Bonus: how $\vec{F}$, U , $\vec{E}$, V connect might appear as an extra-credit problem only

Force $\leftrightarrow$ energy

Force is the (negative) slope of U	$F_x = -\frac{dU}{dx}, \quad \vec{F} = -\vec{\nabla}U$
U is (minus) the work integral of $\vec{F}$ (same statement as $W_{\text{elec}} = -\Delta U$ above, written as a line integral)	$U_f - U_i = -\int_i^f \vec{F} \cdot d\vec{s}$

Field $\leftrightarrow$ potential

Field is the (negative) slope of V	$E_x = -\frac{dV}{dx}, \quad \vec{E} = -\vec{\nabla}V$
V is (minus) the line integral of $\vec{E}$	$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}$

The two rows are the same relationship twice: U and V are what you integrate a force/field *against* to get; $\vec{F}$ and $\vec{E}$ are what falls out when you differentiate U or V . Swap $U \rightarrow V$ and $\vec{F} \rightarrow \vec{E}$ (divide through by q) and the left column becomes the right column.

$\vec{\nabla} \equiv \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}$ — the gradient operator, in Cartesian coordinates.

Companion simulations at hoppese.github.io/Intro-physics-sims/: electroscope, e-fields-conductors-and-insulators, electrostatics-explorer, field-superposition-explorer, discrete-charge-builder-3d, gauss-flux, gauss-applications, assembling-charges, pe-vs-separation, equipotential-explorer.